Lie algebras

Definition: A Lie algebra is an algebra with product $[xy]$ satisfying
- $[xx]=0$ (skew symmetry),
- $[x[yz]]+[y[zx]]+[z[xy]]=0$ (Jacobi identity).

Examples:
- $gl(n,F)$, the space of matrices with $[AB]=AB-BA$;
- $sl(n,F)=\{A\in gl(n,F): \operatorname{Tr} A=0\}$;
- $o(n,F)=\{A\in gl(n,F): A^T+A=0\}$;
- $t(n,F)$, $u(n,F)$, the spaces of upper, resp. strictly upper, triangular matrices.

Def: The centre of a Lie algebra L is $Z(L)=\{x\in L: [xl]=0\ \forall l\in L\}$

Definition: The adjoint representation of L is the map $L\xrightarrow{ad} gl(L)$ given by

$$ad\,x: y\longmapsto [xy].$$

Note that ad is a Lie algebra homomorphism, and that $\operatorname{Ker}(ad)=Z(L)$.

Definition: Let A be an algebra. A linear map $D: A\longrightarrow A$ is a derivation if

$$D(xy)=x(Dy)+(Dx)y.$$

We let $\operatorname{Der} A\subseteq gl(A)$ denote the subspace of derivations.
Note that $ad\,x\in \operatorname{Der} L$ if L is a Lie algebra. In other words, for Lie algebras,

$$ad\,L\subseteq \operatorname{Der} L\subseteq gl(L).$$

Definition: A Lie algebra L is simple if it has dimension >1 and has no proper non-trivial ideals.

Definition: Let

$$L^{(1)}=L'=[L,L],$$
$$\vdots$$
$$L^{(i)}=[L^{(i-1)},L^{(i-1)}].$$

We call the sequence $L\supseteq L^{(1)}\supseteq\ldots$ the derived series of L. Note that $L^{(i)}/L^{(i-1)}$ is abelian.

Definition: We call L soluble if $L^{(m)}=0$ for some m.

Examples: $t(n,F)$, $u(n,F)$ are soluble.

Prop: Let L be a Lie algebra.
(i) If L is soluble then so are its subalgebras and quotients.
(ii) If $I\trianglelefteq L$ is an ideal and I, L/I are soluble then so is L.
(iii) If $I,J\trianglelefteq L$ are soluble ideals then so is $I+J$.

Proof:
(i)+(ii) $(L/I)^{(i)}=L^{(i)}+I/I$.
(iii) $I+J/I\cong J/I\cap J$.

$\square$

Definition: The radical of L, denoted $\operatorname{Rad}(L)$, is the (unique) maximal soluble ideal of L.

Definition: We call L semisimple if $\operatorname{Rad}(L)=0$.

Definition: Let

$$L^1=L'=[L,L]$$
$$\vdots$$
$$L^i=[L,L^{(i-1)}]$$

We call the sequence $L\supseteq L^1\supseteq\ldots$ the lower central series (LCS) of L. Note that

$$L^i/L^{i+1}\subseteq Z(L/L^{i+1}).$$

Definition: We call L nilpotent if $L^m=0$ for some m.

Example: $u(n,F)$ is nilpotent, $t(n,F)$ is not.

Theorem (Engel): Let $L \subseteq \mathfrak{gl}(V)$ be a Lie subalgebra consisting of nilpotent matrices. Then V has a basis B wrt which

$$L = \{[T]_B : T \in L\} \subseteq \mathfrak{n}(n, F).$$

In particular, L is a nilpotent Lie algebra.

Theorem (Engel II): A Lie algebra L is nilpotent iff $\mathrm{ad}\,x : L \longrightarrow L$ is nilpotent $\forall x \in L$.

Proof: $\Rightarrow$ clear

$\Leftarrow$ Apply Engel's theorem to the image of $\mathrm{ad} : L \longrightarrow \mathfrak{gl}(L)$. Its kernel is $Z(L)$.

$\square$

Theorem (Lie): Let $L \subseteq \mathfrak{gl}(V)$ be soluble. Then V has a basis wrt which

$$L = \{[T]_B : T \in L\} \subseteq \mathfrak{t}(n, \mathbb{C}).$$

Definition: A representation of a Lie algebra L is a Lie homomorphism $L \xrightarrow{\varrho} \mathfrak{gl}(V)$.
It is faithful if $\mathrm{Ker}\,\varrho = 0$.

Prop: Let L be soluble. Then any L-module V has a 1-dimensional submodule.
Proof: Apply Lie's theorem to the image of L in $\mathfrak{gl}(V)$.

Definition: An L-module $V \neq 0$ is irreducible if its only submodules are $0, V$.

Representations of $\mathfrak{sl}(2, \mathbb{C})$.

Let $L = \mathfrak{sl}(2, \mathbb{C})$ with basis $\{e, f, h\}$ with

$$[ef] = h, \ [he] = 2e, \ [hf] = -2f.$$

We will show that L has an irreducible representation in each dimension.

Construction: Let $V = \mathbb{C}[X, Y]$. Let

$$V_d = \mathrm{Span}\{X^d, X^{d-1}Y, \dots, Y^d\} \subseteq V$$

be the space of homogeneous polynomials of degree d. Define $\varphi : L \longrightarrow V_d$ by

$$\varphi(e) = X\frac{\partial}{\partial Y}, \quad \varphi(f) = Y\frac{\partial}{\partial x}, \quad \varphi(h) = X\frac{\partial}{\partial y} - Y\frac{\partial}{\partial x}.$$

In terms of the monomial basis

$$\varphi(e) = \begin{pmatrix} 0 & 1 & & \\ & \ddots & \ddots & \\ & & \ddots & d \\ & & & 0 \end{pmatrix}, \quad \varphi(f) = \begin{pmatrix} 0 & & & \\ d & \ddots & & \\ & \ddots & \ddots & \\ & & 1 & 0 \end{pmatrix}, \quad \varphi(h) = \begin{pmatrix} d & & & \\ & d-2 & & \\ & & \ddots & \\ & & & -d+2 & \\ & & & & -d \end{pmatrix}$$

Note that this is a representation of L.

Prop: V_d is an irreducible L-module.
Proof: Any submodule contains an evector for $\varphi(h)$. But these are the monomials, so applying $\varphi(e)$ and $\varphi(f)$ yields V_d.

$\square$

Lemma: Let V be an L-module. Then

- if $v \in V$ is such that $h \cdot v = \lambda v$ then
  $$h \cdot (e \cdot v) = (\lambda + 2) e \cdot v, \quad h \cdot (f \cdot v) = (\lambda - 2) f \cdot v,$$
- $\exists w \in V \setminus 0$ evector for h, such that $e \cdot w = 0$.

Proof: Firstly,

$$h(ev) = e(hv) + [he]v = \lambda ev + 2ev = (\lambda + 2)ev$$

and similarly for f. Secondly, h has an evector v. Then

$$v, ev, e^2 v, \dots$$

have distinct evalues and are hence linearly independent, until $e^{m+1}v$ is zero. Take $w = e^m v$.

$\square$

Note: Using this, one can show that any irreducible L-module is isomorphic to V_d for some d.

Definition: The Killing form of a Lie algebra L is the map $K : L \times L \longrightarrow \mathbb{C}$ given by

$$(x, y) \longmapsto \mathrm{Tr}(\mathrm{ad}\,x \circ \mathrm{ad}\,y) = \mathrm{Tr}(z \longmapsto [x[yz]])$$

Thm (Jordan normal form): Let $f: V \longrightarrow V$ be linear. Then f can be written uniquely as $f = \partial + u$, where $\partial: V \longrightarrow V$ is diagonalisable, $u: V \longrightarrow V$ nilpotent and $[\partial u]=0$.

Note: If $x: V \longrightarrow V$ has Jordan decomposition $x = \partial + u$ then $ad\,x$ has Jordan decomposition $ad\,x = ad(\partial) + ad(u)$.

Thm (Cartan I): A Lie algebra is soluble iff $K(x,y)=0 \;\forall x \in L, y \in L'$.
Idea of proof: For $\Rightarrow$, use Lie's theorem. For $\Leftarrow$, we first assume $L \leq gl(V)$ and show that $Tr(xy)=0 \;\forall x,y \in L$ implies solubility. We apply this to $ad\,L' \cong L^{(2)}$.
To show the claim, we show that every $x \in L'$ is nilpotent and apply Engel's theorem.
This can be done by showing that the Jordan decomposition of x is $x = 0 + \partial$.

Lemma: Let $I \trianglelefteq L$ with Killing form K_I. Then

(1) $K_I = K|_I$

(2) $I^\perp \trianglelefteq L$ ← wrt K

Proof: The first one is simple. For the second, let $x \in I^\perp$, $y \in L$, $i \in I$. Then
$$K([xy]i) = Tr(ad[xy]\,ad\,i) - Tr(ad\,x\,\underbrace{ad[yi]}_{\in I}) = 0. \qquad \square$$

Theorem (Cartan II): A Lie algebra L is semisimple iff its Killing form is non-degenerate.
Proof: Since $L^\perp$ is an ideal, which is soluble by Cartan I, non-degeneracy is immediate.
Conversely, assume $R = Rad\,L \neq 0$. Then L has an abelian ideal A. We show that $A \leq L^\perp$, ie that $Tr(ad(a)\,ad\,x) = 0 \;\forall x \in L$. This follows from the nilpotency of $ad(a)\,ad\,x$:
$$\ell \xmapsto{ad(a)} \underbrace{[a\ell]}_{\in A} \xmapsto{ad\,x} \underbrace{[x[a\ell]]}_{\in A \text{ since } A \text{ is an ideal}} \xmapsto{ad(a)} 0 \quad \leftarrow A \text{ abelian} \qquad \square$$

Thm: A Lie algebra L is semisimple iff it is the direct sum of simple ideals.
Proof: We only need to show that $I \trianglelefteq L \Rightarrow L = I \oplus I^\perp$. From Cartan I, we know that $I \cap I^\perp = 0$, and from Cartan II, it follows that $L = I \oplus I^\perp$.
← $I, I^\perp$ are semisimple by restricting K
For the converse, note that $[I_i, I_j] \subseteq I_i \cap I_j$ $\square$

Classification of simple Lie algebras.

The strategy to understand semisimple Lie algebras is as follows: ($ad\,x$ diagonalisable)

- Find an abelian subalgebra H consisting of semisimple elements;
- Decompose L into a direct sum of weight spaces for $ad\,H$;
- Use the decomposition to pin down the Lie brackets.

Note that since the elements of $ad\,H$ are diagonalisable, and H is abelian, they are simultaneously diagonalisable. The weight $\alpha \in H^*$ corresponding to a mutual eigenvector $x \in L$ is the map $\alpha: H \longrightarrow \mathbb{C}$ such that
$$ad\,h(x) = \alpha(h)x,$$
and its weight space is
$$L_\alpha = \{x \in L : [hx] = \alpha(h)x \;\forall h \in H\}.$$
Let
$$\Phi = \{\alpha \in H^* \setminus 0 : L_\alpha \neq 0\}.$$
Then
$$L = L_0 \oplus \bigoplus_{\alpha \in \Phi} L_\alpha.$$
← we will want H maximal so that $L_0 = H$.

Prop: Let $\alpha, \beta \in H^*$. Then

(1) $[L_\alpha, L_\beta] \subseteq L_{\alpha+\beta}$;

(2) $\alpha+\beta \neq 0 \Rightarrow K(L_\alpha, L_\beta) = 0$;

(3) L semisimple $\Rightarrow K|_{L_0}$ non-degenerate.

Proof:

(1) This follows since ad$h$ is a derivation;

(2) Let $x \in L_\alpha$, $y \in L_\beta$. Then

$$\alpha(h) K(x,y) = K([hx],y) = -K(x,[hy]) = -\beta(h);$$

(3) Let $y \in L_0 \cap L_0^\perp$, $x = x_0 + \sum_{\alpha \in \Phi} x_\alpha \in L$. By (2), $[xy] = 0$. So y spans an abelian ideal.

$\square$

Note: It follows that for $\alpha \in \Phi$, $x \in L_\alpha$, ad$x$ is nilpotent.

Definition: A Cartan subalgebra is a maximal abelian subalgebra consisting of semisimple elements.

Note: The existence of such an algebra follows from the existence of a semisimple element. This in turn follows from a Jordan decomposition in semisimple Lie algebras and Engel's theorem.

Thm: Let L be semisimple, H a subalgebra consisting of semisimple elements. Then H is a Cartan subalgebra iff $H = C_L(H)$. $\leftarrow \{\ell \in L : [\ell h] = 0 \ \forall h \in H\}$

From now on, L will be a semisimple Lie algebra (over $\mathbb{C}$).

Prop: Let $\alpha \in \Phi$. Then $-\alpha \in \Phi$ and for $x \in L_\alpha^{\setminus 0}$, $\exists y \in L_{-\alpha}$ such that $Sp\{x, y, [xy]\}$ is a Lie subalgebra isomorphic to $sl(2)$, which we call $sl(\alpha)$.

Proof: Let $0 \neq x \in L_\alpha$. Let $y \in L$ such that $K(x,y) \neq 0$. By a previous lemma, the projection of y to $L_{-\alpha}$ will still satisfy $K(x,y) \neq 0$ So $-\alpha \in \Phi$.

Let $\alpha(u) \neq 0$. Then

$$K(u,[xy]) = K([ux],y) = \alpha(u) K(x,y) \neq 0$$

Hence, $[xy] \neq 0$. The fact that $L = Sp\{x, y, [xy]\} \cong sl(2)$ follows from the fact that $L' = L$ and the classification of 3-dimensional Lie algebras.

$\square$

Definition: Let $\alpha \in \Phi$, $\beta \in \Phi \cup \{0\}$. The α-string through β is the set

$$\{\beta + c\alpha : c \in \mathbb{Z}\} \cap \Phi \subseteq \Phi$$

Prop: The subspace

$$M = \sum_{\substack{c \in \mathbb{Z} \\ \beta + c\alpha \in \Phi}} L_{\beta + c\alpha} \subseteq L$$

is an $sl(\alpha)$-submodule.

Prop: Let $\alpha \in \Phi$. Then

- $\dim L_\alpha = 1$
- $n\alpha \in \Phi \Rightarrow n = \pm 1$ if $n \in \mathbb{Z} \setminus 0$

Proof: Let $W = Sp\{H, e_{-\alpha}, L_{n\alpha}\}_{n \in \mathbb{Z}^+}$.

Then $sl(\alpha) = Sp\{e_{-\alpha}, e_\alpha, h = [e_\alpha, e_{-\alpha}]\} \circlearrowright W$. Note that as a commutator, $Tr(\text{ad} h_\alpha|_W) = 0$.

But on W, $$\text{ad} h_\alpha|_W = 0 \oplus \underbrace{-\alpha(h_\alpha)}_{2} \oplus \bigoplus_{n \in \mathbb{Z}^+} n\alpha(h_\alpha) I_{\dim L_{n\alpha}}.$$

Hence, $$2 = \bigoplus_{n \in \mathbb{Z}^+} 2n \dim L_{n\alpha}$$ and the claims follow.

$\square$

Note: We know that on the irreducible representations, $h \in sl(2, \mathbb{C})$ has integral evectors. Taking a decomposition series, it follows that this is true for any representation.

Prop: Let $\alpha, \beta \in \Phi$, $\alpha \pm \beta \neq 0$. Then

1) $\beta(h_\alpha) \in \mathbb{Z}$;
2) The α-root string through β is
   $$\beta - r\alpha, \beta-(r-1)\alpha, \dots, \beta + q\alpha$$
   where $q, r \geq 0$, $\beta(h_\alpha) = r - q$;
3) If $\alpha + \beta \in \Phi$ then
   $$[e_\alpha e_\beta] = \lambda e_{\alpha+\beta}$$
   where $\lambda \neq 0$;
4) $\beta - \beta(h_\alpha)\alpha \in \Phi$. ← this follows from (2)

Proof: The first claim follows from the above note. Let $M = \sum_{c \in \mathbb{Z}} \overbrace{L_{\beta+c\alpha}}^{\text{1-dim espaces for ad} h_\alpha}$. This is a $sl(\alpha)$-module. On $L_{\beta+c\alpha}$, adh_α has evalue $(\beta + c\alpha)(h_\alpha) = \beta(h_\alpha) + 2c$, so its evalues on M are all either even or odd. Hence, by the argument via decomposition series, since the evalues have multiplicity one, M is irreducible. So $M \cong V_\lambda$ and

$$\{\beta(h_\alpha) + 2c : \beta + c\alpha \in \Phi\} = \{-\lambda, -\lambda+2, \dots, \lambda\}$$

Letting $\lambda = \beta(h_\alpha) + 2q$, $-\lambda = \beta(h_\alpha) - 2r$, we get the second claim, where $r, q \geq 0$ follows from $\beta(h_\alpha) - \beta(h_\alpha) + 2 \cdot 0 \in \{-\lambda, \dots, \lambda\}$.

Finally, e_α acts on $M \cong V_\lambda$ as $\begin{pmatrix} 0 & 1 & & \\ & \ddots & \ddots & \\ & & \ddots & \lambda \\ & & & 0 \end{pmatrix}$, so $[e_\alpha e_\beta] = 0$ implies that $[h_\alpha e_\beta] = \lambda e_\beta$, so that $\beta(h_\alpha) = \lambda$, ie $q = 0$. But $\alpha + \beta \in \Phi$, so $q \geq 1$. □

Note: If $H \subseteq L$ is a Cartan subalgebra, then for $h_1, h_2 \in H$,

$$K(h_1, h_2) = \sum_{\alpha \in \Phi} \alpha(h_1)\alpha(h_2).$$

Prop: $\overset{W}{\overset{\shortparallel}{\text{Span}(\Phi)}} = H^*$.

Proof: If not, let $h \in \text{Ann}_H W$ so $\alpha(h) = 0$ $\forall \alpha \in \Phi$. Hence, by the previous note, $K(h, H) = 0$. Since $K(h, L_\alpha) = 0$, $h \in L^\perp = 0$. □

Notation: Let $\theta : H \xrightarrow{\cong} H^*$ denote the map $h \longmapsto \theta_h = K(h, \cdot)$.

$K|_H$ is non-degenerate

Denote its inverse by $H^* \ni \alpha \longmapsto t_\alpha \in H$, so that $\theta_{t_\alpha} = \alpha$ and $t_{\theta_h} = h$.

Prop: Let $\alpha \in \Phi$, $x \in L_\alpha$, $y \in L_{-\alpha}$. Then $[xy] = K(x,y)t_\alpha$.

Proof: For $h \in H$, $K(h, [xy]) = K([hx], y) = \alpha(h)K(x,y)$

$$= K(t_\alpha, h)K(x,y) \overset{\text{symmetry of } K}{=} K(h, K(x,y)t_\alpha),$$

so the claim follows from non-degeneracy of $K|_H$. □

Prop:
$$t_\alpha = \frac{h_\alpha}{K(e_\alpha, e_{-\alpha})}, \quad h_\alpha = \frac{2t_\alpha}{K(t_\alpha, t_\alpha)};$$

$$K(t_\alpha, t_\alpha)K(h_\alpha, h_\alpha) = 4.$$

Proof: The first identity follows from above, the second from plugging in
$$\alpha(h_\alpha) = 2 = K(e_\alpha, e_{-\alpha})\alpha(t_\alpha) = K(e_\alpha, e_{-\alpha})K(t_\alpha, t_\alpha).$$

Finally,
$$K(h_\alpha, h_\alpha) = K\left(\frac{2t_\alpha}{K(t_\alpha,t_\alpha)}, \frac{2t_\alpha}{K(t_\alpha,t_\alpha)}\right) = \frac{4}{K(t_\alpha,t_\alpha)^2}K(t_\alpha, t_\alpha).$$ □

Corollary: $K(h_\alpha, h_\beta) \in \mathbb{Z}$, $K(t_\alpha, t_\beta) \in \mathbb{Q}$.

Proof: We have
$$K(h_\alpha, h_\beta) = \sum_\Phi \gamma(h_\alpha)\gamma(h_\beta) \in \mathbb{Z},$$
so the claim follows from above. □

Prop: Let $E = \mathrm{Span}_{\mathbb{R}} \Phi$. Let $(\theta_{h_1}, \theta_{h_2}) := K(h_1, h_2)$. Then
- $(\cdot,\cdot) : E \times E \longrightarrow \mathbb{R}$;
- $(\cdot,\cdot)$ is positive definite.

Proof: On Φ, $(\cdot,\cdot)$ is $\mathbb{Q}$-valued, so the first claim follows. For the second,

$$(\theta_h, \theta_h) = K(h,h) = \sum_{\Phi} \gamma(h)^2 = \sum_{\Phi} K(t_\gamma, h)^2 = \sum_{\Phi} (\underbrace{\gamma, \theta_h}_{\in \mathbb{R}})^2 . \qquad \square$$

DOES THIS MEAN THAT $h \in H, \gamma \in \Phi \Rightarrow \gamma(h) \in \mathbb{R}$??

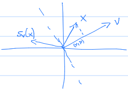
Root systems

Notation: If E is an inner product space, we define the reflections

$$s_v : E \longrightarrow E, \quad x \longmapsto x - \frac{2(v,x)}{(v,v)} v .$$

This is an isometry. We also denote

$$\langle x, v \rangle = \frac{2(x,v)}{(v,v)} .$$



Definition: A subset $R \subseteq E$ is a root system if
1. $|R| < \infty$, $0 \notin R$, $\mathrm{Span}\, R = E$;
2. $\alpha, \lambda\alpha \in R \Rightarrow \lambda = \pm 1$;
3. $\alpha \in R \Rightarrow s_\alpha(R) = R$; $\longleftarrow$ in particular, $-v \in R$ if $v \in R$
4. $\alpha, \beta \in R \Rightarrow \langle \beta, \alpha \rangle \in \mathbb{Z}$.

Its rank is $\dim E$.

Prop: $\Phi \subseteq E$ is a root system.

Proof:
1. Clear;
2. We proved this for $\lambda \in \mathbb{Z}$, the general case is probably similar;

3.,4. We have $\mathbb{Z} \ni \beta(h_\alpha) = K(t_\beta, h_\alpha) = K\left(t_\beta, \frac{2t_\alpha}{K(t_\alpha, t_\alpha)}\right) = \frac{2(\beta,\alpha)}{(\alpha,\alpha)} = \langle \beta, \alpha \rangle$,

so this follows from a previous proposition, since $s_\alpha(\beta) = \beta - \langle \beta, \alpha \rangle \alpha$.

Example ($L = \mathfrak{sl}(n, \mathbb{C})$): We have $H = \left\{ \begin{pmatrix} * & & \\ & \ddots & \\ & & * \end{pmatrix} \right\}$, with root spaces $\mathrm{Sp}\{E_{ij}\}_{i \neq j}$ and roots $\varepsilon_i - \varepsilon_j$ $(i \neq j)$. So

$$E = \mathrm{Sp}\, \Phi = \left\{ \sum \lambda_i \varepsilon_i : \sum \lambda_i = 0 \right\} \qquad \boxed{A_{n-1}}$$

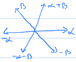
After rescaling, the inner product is $(\varepsilon_i, \varepsilon_j) = \delta_{ij}$.

Example (Rank 2 root systems):

(a) $\mathfrak{sl}(3, \mathbb{C})$: $\alpha = \varepsilon_1 - \varepsilon_2$, $\beta = \varepsilon_2 - \varepsilon_3 \rightsquigarrow \Phi = \{ \pm\alpha, \pm\beta, \pm(\alpha+\beta) \}$

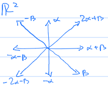
$(\alpha,\beta) = -1$, $(\alpha,\alpha) = (\beta,\beta) = 2$

$\rightsquigarrow$

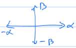


(b) $\alpha = e_2$, $\beta = e_1 - e_2 \in \mathbb{R}^2$

$\rightsquigarrow$



(c)



Definition: A root system R is reducible if $R = R_1 \cup R_2$ with $(R_1, R_2) = 0$.

It can be shown that R irreducible $\Rightarrow L$ simple. Also, by showing first that $\mathrm{Aut}\, L$ acts transitively on Cartan subalgebras, it can be shown that any two Cartan subalgebras give the same root system.

Thm: Let Φ be a root system. Then there is a unique semisimple Lie algebra L corresponding to it.

Idea: For uniqueness, we get the structure constants from Φ: Let $\alpha, \beta \in \Phi$. From the α-string through β

$$\beta - r\alpha, \dots, \beta + q\alpha,$$

we get

$$[h_\alpha, h_\beta] = 0, \ [h_\alpha, e_\alpha] = \beta(h_\alpha) e_\beta = (r-q) e_\beta$$
$$[e_\alpha, e_{-\alpha}] = h_\alpha, \ [e_\alpha, e_\beta] = 0 \text{ if } \alpha+\beta \notin \Phi \cup \{0\}$$
$$[e_\alpha, e_\beta] = \pm(q+1) e_{\alpha+\beta} \text{ if } \alpha + \beta \in \Phi$$

$\square$

Definition: The Weyl group of $R \subseteq E$ is the group

$$W(R) = \langle s_\alpha : \alpha \in R \rangle \leq GL(E).$$

Note that $W(R) \circlearrowright R$ faithfully since $\operatorname{Sp} R = E$. In particular, $W(R)$ is finite.

Definition: A subset $B \subseteq R$ is a base of R if
- B is a basis for E,
- $\forall \beta \in R$, $\beta \in \mathbb{Z}_{\geq 0} \cdot B \cup \mathbb{Z}_{\leq 0} \cdot B$.

If $\beta \in \mathbb{Z}_{\geq 0} \cdot B$ then β is called a positive root (wrt B), and negative otherwise.

Thm: Every root system has a base. $W(R) \circlearrowright$ bases sharply 1-transitively.

Definition: The Dynkin diagram $\Delta(R)$ is a graph with vertices B and $d_{\alpha\beta}$ edges between $\alpha, \beta \in B$, where

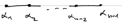
$$d_{\alpha\beta} = \langle \beta, \alpha \rangle \langle \alpha, \beta \rangle = \frac{4(\alpha,\beta)^2}{(\alpha,\alpha)(\beta,\beta)} \in \{0,1,2,3\}.$$

If $d_{\alpha\beta} > 1$, draw an arrow from the longer vertex to the shorter one.

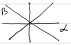
Examples:
- A_{n-1}: roots $\varepsilon_i - \varepsilon_j$, base $\overbrace{\varepsilon_1 - \varepsilon_2}^{\alpha_1}, \overbrace{\varepsilon_2 - \varepsilon_3}^{\alpha_2}, \dots, \overbrace{\varepsilon_{n-1} - \varepsilon_n}^{\alpha_{n-1}}$,

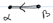
  $d_{\alpha_i \alpha_{i+1}} = 1, \ d_{\alpha_i \alpha_j} = 0 \ j \neq i \pm 1$

  $\Delta(A_{n-1})$:

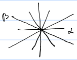


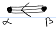
- B_2: base α, β





- G_2: base α, β





Prop: A root system R is irreducible iff $\Delta(R)$ is connected.

We want to show that the classical Lie algebras

$$\mathfrak{sl}(n,\mathbb{C}), \ \mathfrak{so}(n,\mathbb{C}), \ \mathfrak{sp}(2n,\mathbb{C})$$

are simple, except $\mathfrak{so}(2,\mathbb{C})$, $\mathfrak{so}(4,\mathbb{C})$, $\mathfrak{sl}(1,\mathbb{C})$, $\mathfrak{so}(1,\mathbb{C})$. Using Dynkin diagrams, this will follow from the next proposition.

Prop: Let $Z(L) = 0$, H a subalgebra with weight decomposition

$$L = H \oplus \bigoplus_{\Phi} L_\alpha$$

where $\Phi \subseteq H^* \setminus 0$. Suppose that

I) $\dim L_\alpha = 1 \quad \forall \alpha \in \Phi$,

II) $\alpha \in \Phi \Rightarrow -\alpha \in \Phi$,

III) $[[L_\alpha, L_{-\alpha}], L_\alpha] \neq 0 \quad \forall \alpha \in \Phi$.

Then L is semisimple.

Proof: Let $I \leq L$ be abelian. Since $\operatorname{ad} H : I \to I$,

$$I = (I \cap H) \oplus \bigoplus_{\alpha \in \Phi} (I \cap L_\alpha)$$

If $I \cap L_\alpha \neq 0$ then $L_\alpha \subseteq I$, so

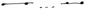
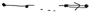
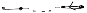
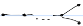
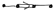
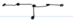
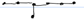
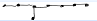
$$0 \neq [[L_\alpha, L_{-\alpha}], L_\alpha] \subseteq [[I, L_\alpha], I] \subseteq [I, I] = 0.$$

Hence, $I \subseteq H$. Since $Z(L) \neq 0$, $[I, L_\alpha] \neq 0$ for some $\alpha \in \Phi$. But

$$[I, L_\alpha] \subseteq I \cap [H, L_\alpha] \subseteq I \cap L_\alpha = 0.$$ $\square$

Prop: $\Delta(R) \cong \Delta(R') \Longrightarrow R \cong R'$.

Thm: The Dynkin Diagrams for irr. root systems are

	A_n $(n \geq 1)$	$\mathfrak{sl}(n+1, \mathbb{C})$
	B_n $(n \geq 2)$	$\mathfrak{so}(2n+1, \mathbb{C})$
	C_n $(n \geq 2)$	$\mathfrak{sp}(2n, \mathbb{C})$
	D_n $(n \geq 3)$	$\mathfrak{so}(2n, \mathbb{C})$
	G_2	$\mathfrak{g}_2$
	F_2	$\mathfrak{f}_2$
	E_6	$\mathfrak{e}_6$
	E_7	$\mathfrak{e}_7$
	E_8	$\mathfrak{e}_8$